

- Table 10.1 shows the potential difference across the neurone membrane at three sequential time points, **X**, **Y** and **Z**, just before and during an action potential.

time point	potential difference / mV
X	-70
Y	+40
Z	-90

- [7]



- (b) Suggest how one action potential causes an action potential in an adjacent section of the axon of an unmyelinated neurone.

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..... [2]

- (c) Vertebrate animals generally have a myelin sheath around the axons of motor neurones.

Explain why the presence of a myelin sheath around a motor neurone axon is an advantage.

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..... [3]

[Total: 12]